End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
(Course Name:B.Tech.) (Semester:2nd)
(May, 2024)

Subject Code: BAI 106	Subject: Database Management Systems

Time :3 Hours — Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(4*5=20)	
	a) Name any four type of attributes in ER Diagram with examples and its representation.		
	b) What do you mean by generalization and specialization?		
	c) Explain serial and concurrent schedule with suitable examples.		
	d) Explain committ and rollback command with suitable examples.		
	e) What is the difference between procedures and functions?		
	UNIT-I		
Q2	a) Write short notes on Heirarchical and Network data models. b)Explain the key distinctions between UPDATE vs. ALTER and DELETE vs. DROP in SQL? Include syntax and examples to illustrate.	(5,5)	
Q3	a) Draw and explain the detailed system architecture of DBMS. b) Explain advantages of Database management system over file processing systems.	(5,5)	
	UNIT-II		
Q4	a) Let E1 and E2 be two entities in an E/R diagram with simple single-valued attributes. R1 and R2 are two relationships between E1 and E2, where R1 is one- to-many and R2 is many-to-many. R1 and R2 do not have any attributes of their own. Calculate the minimum number of tables required to represent this situation in the relational model? b) Draw an ER diagram for the following application from the manufacturing industry: i. Each supplier has a unique name. ii. More than one supplier can be located in the same city. iii. Each part has a unique part number. iv. Each part has a colour. v. A supplier can supply more than one part. vi. A part can be supplied by more than one supplier. vii. A supplier can supply a fixed quantity of each part.	(3,7)	

Q5	Consider the following schema: Suppliers (sid : integer, sname : string, address : string) Parts (pid : integer, pname : string, color : string) Catalog (sid : integer, pid : integer, cost : real) The key fields are underlined and domain of each field is listed after the field name a) write following SQL queries i. Find the names of suppliers who supply more than one red part. ii. Calculate the average difference between the cost of parts offered by a specific supplier (use your own 'sname' value) and the overall average cost for all parts. iii. List the most expensive part from each supplier, including the part name, color, and cost. iv. Find the names of suppliers that don't have any parts listed in the Catalog table. v. List the names of parts that are supplied by at least two different suppliers. b) Write following relational algebra queries i. Project the supplier names and addresses of those suppliers who provide parts with a cost less than $50. ii. Project the part names and colors of those parts supplied by suppliers located in a specific city. iii. Find the names of suppliers who provide parts with a cost less than $50 and are located in a specific city. iv. Project the supplier names and addresses along with the names of parts they provide. v. Find the names of suppliers who provide only parts of a specific color.	(5,5)	
	UNIT-III		
Q6	a. What is Transaction? Explain ACID properties of transaction. b. Discuss conflict serialiazability and 2PL protocol. Explain with the help of example occurrence of deadlock in DBMS and measures to prevent it.	(5,5)	
Q7	a) What is normalization? Explain first, second, third Normal forms with suitable example. b) Consider a relation R with attribute set (A, B, C, D,E) and functional dependency set F = (AB→ C, B → D, D→ E).Decompose the above Relation R into 3NF form. Check if the decomposition is lossless and dependency preserving or not. Discuss	(5,5)	
	UNIT-IV		
Q8	a)What is database trigger? Explain any two types of trigger. b)Explain the differences between OODBMS and ORDBMS.	(5,5)	
Q9	Explain all the related steps of Querry Processing in DBMS with neat diagram.	(10)	